

AYUSH GIRI

+91 9569285853 | ayushgiri2125@gmail.com | [GitHub](#) | [LinkedIn](#) | [LeetCode](#) | [Portfolio](#)

Education

SRM University AP, Amaravati, India
B.Tech in Computer Science & Engineering

2023 – 2027
CGPA: 8.35/10

Technical Skills

Core: Python, Java, C++, JavaScript, TypeScript, SQL

Full-Stack: React, Next.js, FastAPI, Node.js, Express.js, RESTful API Design, JWT Authentication

Databases: PostgreSQL, MongoDB, SQLAlchemy (ORM)

Infra & Tooling: Docker, Redis, Celery, Socket.IO, GitHub Actions (CI), pytest

AI/LLM: Google Gemini API, Prompt Engineering, Schema-Validated LLM Pipelines

Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Research Experience

Research Intern

June 2025 - Aug 2025

SRM University AP

- Built distributed computation workflows and feature extraction frameworks to improve machine learning experimentation efficiency.
- Automated and tracked **50+** experiments through reproducibility-first logging infrastructure built into the workflow pipeline.
- Re-engineered feature extraction pipelines for CIFAR-10 image classification, improving model accuracy by **8%**.

Projects

Nexora : AI-Powered Networking CRM Platform

[Link](#)

React 19 • Vite • Tailwind CSS • FastAPI • PostgreSQL • SQLAlchemy • Celery • Redis • Docker

- Architected role-based access control with rate limiting across **310+** REST endpoints, supporting dual JWT and Supabase authentication modes.
- 100% environment parity across **dev/prod**, achieved by containerizing the full stack (React, FastAPI, PostgreSQL, Redis) with Docker Compose.
- Automated regression coverage on all critical auth/API paths, achieved by building a CI/CD-ready pytest + Playwright pipeline.

Aether : Multi-Agent React UI Generation System

[Link](#)

Next.js 14 • TypeScript • React • Zod • Google Gemini API

- Enforced schema-validated outputs across **5** API routes using **Zod**, reducing malformed LLM responses.
- Added a custom prompt-injection guard to sanitize LLM-facing input before it reaches the model.
- Built a mock-mode fallback enabling fully offline development and demos during live API outages.

CodeForge : Real-Time Collaborative Code Editor

[Link](#)

React • Node.js • Express.js • MongoDB • Socket.IO • JWT • Judge0 API

- Built a MERN-stack collaborative code editor with real-time multi-user editing and room-based sessions via Socket.IO.
- Less than **2% sandbox escapes** in testing, by integrating secure, isolated code execution via a protected /api/execute endpoint on **Judge0 CE**.
- Implemented JWT-authenticated access control for collaboration rooms, restricting session participation to authorized users.

Certifications

MongoDB Certified Associate Developer MongoDB, Inc.

[Certificate](#)

Schema design, aggregation pipelines, indexing, and CRUD on document-oriented databases.

Oracle Certified Professional: Java SE 17 Developer Oracle

[Certificate](#)

Validated proficiency in core Java, OOP design, exception handling, generics, and concurrency.